DBMS-SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

PROJECT TITLE-FLIPKART E-COMMERCE PLATFORM ANALYSIS

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK  SELLER_ID
----------- ---------- ---------- ----------
       101
Laptop
          1      56000         25        201

       102
Smartphone
          1      25000         30        202

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK  SELLER_ID
----------- ---------- ---------- ----------

       103
T-Shirt
          2        799         50        203

       104
Mixer Grinder

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK  SELLER_ID
----------- ---------- ---------- ----------
          3       2500         15        201

       105
Python Programming Book
          4        650          0        204
```

Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       101
Laptop
     56000         25

       102
Smartphone
     25000         30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       103
T-Shirt
       799         50

       104
Mixer Grinder

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2500         15

       105
Python Programming Book
       650          0
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       103
T-Shirt
       799

       105
Python Programming Book
       650

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

Find products with price greater than 100

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price > 100;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Laptop
     56000

       102
Smartphone
     25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
T-Shirt
       799

       104
Mixer Grinder

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      2500

       105
Python Programming Book
       650
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       105
Python Programming Book
       650

       103
T-Shirt
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       104
Mixer Grinder
      2500

       102
Smartphone

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
     25000

       101
Laptop
     56000
```

Products from highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Laptop
     56000

       102
Smartphone
     25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       104
Mixer Grinder
      2500

       103
T-Shirt

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       799

       105
Python Programming Book
       650
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT c.Category_Name
  2  FROM Product p
  3  JOIN Category c ON p.Category_ID = c.Category_ID
  4  ORDER BY c.Category_Name;

CATEGORY_NAME
--------------------------------------------------
Books
Electronics
Fashion
Home Appliances
```

2. Search Products

2.1 Search Products Based on Price

Find products priced between ₹500 and ₹30,000.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 500 AND 30000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Smartphone
     25000

       103
T-Shirt
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       104
Mixer Grinder
      2500

       105
Python Programming Book

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       650
```

2.2 Search Products Based on Category

Display products in the Electronics category.

```
SQL> SELECT p.Product_ID, p.Product_Name, c.Category_Name, p.Price
  2  FROM Product p
  3  JOIN Category c ON p.Category_ID = c.Category_ID
  4  WHERE c.Category_Name = 'Electronics';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      PRICE
-------------------------------------------------- ----------
       101
Laptop
Electronics                                         56000

       102
Smartphone
Electronics                                         25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      PRICE
-------------------------------------------------- ----------
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT * FROM Customer;

        ID NAME                      AGE ADDRESS                SALARY
---------- -------------------- ---------- -------------------- ----------
         1 Renu                         19 Chennai                 25000
         2 Arjun                        20 Tambaram                30000
         3 Priya                        19 Vandalur                28000
         4 Rahul                        21 Chengalpattu            35000
         5 Keerthana                    20 Guduvanchery            32000
```

Search for a particular customer

```
SQL> SELECT ID, NAME, AGE, ADDRESS
  2  FROM Customer;

        ID NAME                      AGE ADDRESS
---------- -------------------- ---------- --------------------
         1 Renu                         19 Chennai
         2 Arjun                        20 Tambaram
         3 Priya                        19 Vandalur
         4 Rahul                        21 Chengalpattu
         5 Keerthana                    20 Guduvanchery
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM Product
  3  WHERE Price < 30000 AND Stock > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       102
Smartphone
     25000         30

       103
T-Shirt
       799         50

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       104
Mixer Grinder
      2500         15
```

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Category_ID, Price
  2  FROM Product
  3  WHERE Category_ID = 1 OR Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------
       101
Laptop
          1      56000

       102
Smartphone
          1      25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------

       103
T-Shirt
          2        799
```

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Category_ID, Price
  2  FROM Product
  3  WHERE Category_ID IN (1, 3, 5);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------
       101
Laptop
          1      56000

       102
Smartphone
          1      25000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------

       104
Mixer Grinder
          3       2500
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 1000 AND 30000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Smartphone
     25000

       104
Mixer Grinder
      2500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT
  2      COUNT(*) AS Total_Products,
  3      MIN(Price) AS Lowest_Price,
  4      MAX(Price) AS Highest_Price,
  5      ROUND(AVG(Price), 2) AS Average_Price
  6  FROM Product;

TOTAL_PRODUCTS LOWEST_PRICE HIGHEST_PRICE AVERAGE_PRICE
-------------- ------------ ------------- -------------
             5          650         56000       16989.8
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS Customer_Name,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT,
  7      o.ORDER_STATUS
  8  FROM Customer c
  9  JOIN Orders o ON c.ID = o.CUSTOMER_ID
 10  ORDER BY c.ID, o.ORDER_ID;

        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         1 Renu                        401 15-SEP-26        55000
Shipped

         2 Arjun                       402 16-SEP-26        25000
Shipped

         3 Priya                       403 17-SEP-26         1598
Processing


        ID CUSTOMER_NAME          ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- -------------------- ---------- --------- ------------
ORDER_STATUS
------------------------------
         4 Rahul                       404 18-SEP-26         2500
Delivered
```

5.3 Product Order Report

```
SQL> SELECT
  2      o.ORDER_ID,
  3      p.PRODUCT_ID,
  4      p.PRODUCT_NAME,
  5      od.QUANTITY,
  6      od.UNIT_PRICE,
  7      od.QUANTITY * od.UNIT_PRICE AS ITEM_TOTAL
  8  FROM Orders o
  9  JOIN Order_Details od ON o.ORDER_ID = od.ORDER_ID
 10  JOIN Product p ON od.PRODUCT_ID = p.PRODUCT_ID
 11  ORDER BY o.ORDER_ID;

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ----------
       401        101
Laptop
         1      55000      55000

       402        102
Smartphone
         1      25000      25000

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ----------

       403        103
T-Shirt
         2        799       1598

       404        104
Mixer Grinder

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL
---------- ---------- ----------
         1       2500       2500
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS Customer_Name,
  4      COUNT(o.ORDER_ID) AS Total_Orders
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7      ON c.ID = o.CUSTOMER_ID
  8  GROUP BY c.ID, c.NAME
  9  ORDER BY c.ID;

        ID CUSTOMER_NAME        TOTAL_ORDERS
---------- -------------------- ------------
         1 Renu                            1
         2 Arjun                           1
         3 Priya                           1
         4 Rahul                           1
         5 Keerthana                       0
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.ID,
  3      c.NAME AS Customer_Name,
  4      NVL(SUM(o.TOTAL_AMOUNT), 0) AS Total_Purchase_Amount
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7      ON c.ID = o.CUSTOMER_ID
  8  GROUP BY c.ID, c.NAME
  9  ORDER BY c.ID;

        ID CUSTOMER_NAME        TOTAL_PURCHASE_AMOUNT
---------- -------------------- ---------------------
         1 Renu                                55000
         2 Arjun                               25000
         3 Priya                                1598
         4 Rahul                                2500
         5 Keerthana                               0
```